Complex number

複數 $i=\sqrt{-1}$

若 $x=0$, $z=iy_1$ 稱為純虛數

Imaginary

$Im(z)$; $Re(z)$ 實部

e.g $z_1=8+3i$, $z_2=9-2i$ 求: $Re(z_1)$, $Im(z_1)$

z_1+z_2, $z_1\cdot z_2$

$Re(z_1)=8$, $Im(z_1)=3$

$z_1+z_2=17+i$

$z_1\cdot z_2=72+6+27i-16i$

$=78+11i$

除法: 有理化

$z_1=x_1+iy_1$, $z_2=x_2+iy_2$

$$z=\frac{z_2}{z_1}=\frac{x_2+iy_2}{x_1+iy_1}=\frac{(x_2+iy_2)(x_1-iy_1)}{(x_1+iy_1)(x_1-iy_1)}=\frac{(x_2+iy_2)(x_1-iy_1)}{x_1^2+y_1^2}$$

e.g $z_1=8+3i$, $z_2=9-2i$ 求 $\frac{z_1}{z_2}$

$$\frac{8+3i}{9-2i}=\frac{(8+3i)(9+2i)}{81+4}=\frac{72-6+16i+27i}{85}=\frac{66+43i}{85}$$

Complex conjugate numbers (共軛複數)

$z=x+iy$

$\bar{z}=x-iy$

e.g $z_1=4+3i$, $z_2=2+5i$

證 $Im(z_1)=\frac{1}{2i}(z_1-\bar{z}_1)$

$z_1=4+3i$

$\bar{z}_1=4-3i$

$z_1-\bar{z}_1=6i$

$\frac{6i}{2i}=3$

又 $Im(z_1)=3$ $\therefore Im(z_1)=\left(\frac{z_1-\bar{z}_1}{2i}\right)$

Commutation ; 交換率

Associative law; 結合率

Distributive law; 分配率

$\overline{(z_1+z_2)}=\bar{z}_1+\bar{z}_2$

$\overline{(z_1-z_2)}=\bar{z}_1-\bar{z}_2$

$\overline{z_1z_2}=\bar{z}_1\cdot\bar{z}_2$

$\overline{\left(\frac{z_1}{z_2}\right)}=\frac{\bar{z}_1}{\bar{z}_2}$

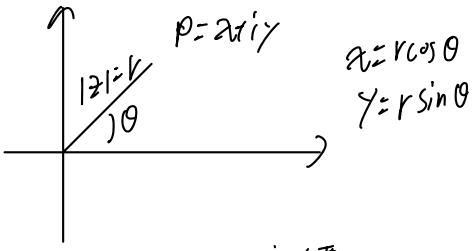
H.W 證

陳宥宜

Polar Form of complex numbers

複數的極座標表示

以極座標 r, θ 表示



$x = r\cos\theta$

$y = r\sin\theta$

$\therefore$ $z = x+iy$ 的極座標

$z = r\cos\theta + ir\sin\theta$

$= r(\cos\theta + i\sin\theta)$

θ 為 z 的 argument，以 $\arg z$ 表示

(輻角)

$\therefore \theta = \arg z$，$\tan\theta = \frac{y}{x}$

θ 單位 (rad)

r 以 $|z|$ 表示

$$|z| = r = \sqrt{x^2+y^2} = \sqrt{z\cdot\bar{z}}$$

當 $z \neq 0$，$\because \sin\theta$ 及 $\cos\theta$ 的週期為 2π

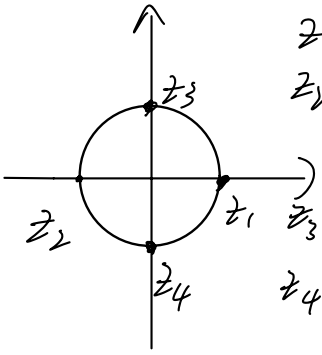
$\therefore$ 2π 範圍中 θ 的 pricipal value

(主值)

以 $\mathrm{Arg}\, z$ 表示

↑

要大寫，$-\pi < \mathrm{Arg}\, z \leq \pi$



$z_1 = 1$ $\mathrm{Arg}\, z = 0$

$z_2 = -1$ $\mathrm{Arg}\, z = \pi$

$z_3 = i$ $\mathrm{Arg}\, z = \frac{\pi}{2}$

$z_4 = -i$ $\mathrm{Arg}\, z = \frac{3\pi}{2}$

$z \neq 0$ 與主值相同的其他解為

$\arg z = \mathrm{Arg}\, z \pm 2n\pi$

e.g. $z = 1+i$

求 polar form，$|z|$，$\mathrm{Arg}\, z$，$\arg z$

$|z| = \sqrt{2} = r$

$\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$

$z = \sqrt{2}\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i\right)$

$= \sqrt{2}\left(\cos\frac{1}{4}\pi + i\sin\frac{1}{4}\pi\right) = \sqrt{2}e^{i\frac{\pi}{4}}$

$\mathrm{Arg}\, z = \frac{\pi}{4}$

$\arg z = \frac{\pi}{4} \pm 2n\pi$

補：利用 Euler formula

$e^{i\theta} = \cos\theta + i\sin\theta$

課本例題

Multiplication & division in polar form
(乘) (除)

設 $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$

$z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$

若 Multiplication

$$z_1 z_2 = r_1 r_2 + r_1(\cos\theta_2 + i\sin\theta_2) + r_2(\cos\theta_1 + i\sin\theta_1) + (\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2)$$

$$= r_1 r_2$$

$|z_1 z_2| = r_1 r_2 = |z_1||z_2|$

若 $z_1 = z_2$

$z^n = r^n[\cos(n\theta) + i\sin(n\theta)] = r^n e^{in\theta}$ #

$\therefore |z_1 z_2| = |z_1||z_2|$

Root (開根號)

若 $z = w^n$

給一個 w 得一個 z

給一個 z 得 n 個 w

$\therefore$ 設 $z = r(\cos\theta + i\sin\theta)$

$w = R(\cos\phi + i\sin\phi)$

$\Rightarrow w^n = R^n(\cos n\phi + i\sin n\phi)$

$\because z = w^n = R^n(\cos n\phi + i\sin n\phi) = r(\cos\theta + i\sin\theta)$

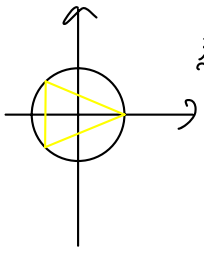
$\therefore n\phi = \theta + 2\pi(k)$ $(k = 1, 2 \cdots$

$\Rightarrow \phi = \frac{\theta}{n} + \frac{2\pi k}{n}$

$\because z \neq 0$

n 個 w 分佈在半徑 $\sqrt[n]{r}$, 圓心在 0 點

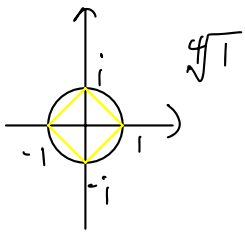
在圓上有 n 個 w 的連線形成 n 邊形



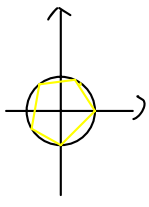
$\sqrt[3]{1} = (1), (-\frac{1}{2}, \frac{i\sqrt{3}}{2}), (-\frac{1}{2}, -\frac{i\sqrt{3}}{2})$

$\sqrt[n]{z} = \sqrt[n]{r}\left(\cos\frac{\theta + 2\pi k}{n} + i\sin\frac{\theta + 2\pi k}{n}\right)$

$k = 0, 1, 2 \cdots (n-1)$



$\sqrt[4]{1}$



$\sqrt[5]{1} = \cos\frac{2\pi k}{5} + i\sin\frac{2\pi k}{5}$

複數函數

(1) f 為定義在 S 的函數
每個對應一個 w

$w = f(z)$

z 為 complex variable 變數

w 為 f 在 z 的值

$w = u + iv$

$u = u(x, y)$

$v = v(x, y)$

$w(x, y) = u(x, y) + i\,v(x, y)$

e.g. $w = f(z) = z^2 + 3z$，求 u & v 和 f 在 $z = 1 + 3i$ 之值

$z = 1 + 3i$

$w = u + iv = (x+yi)^2 + 3(x+yi)$

$= x^2 - y^2 + 2ixy + 3x + 3yi$

$= x^2 - y^2 + 3x + i(2xy + 3y)$

$u = (x^2 - y^2 + 3x)$

$v = (2xy + 3y)$

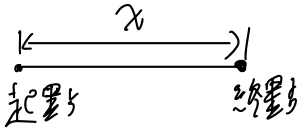
$f(1+3i) = (1+3i)^2 + 3(1+3i)$

$= 1 - 9 + 9 + i(6+9)$

$= 1 + 15i$

向量

Vector：具方向&大小



(x, y, z) 卡式座標

(a, b, c)

$x_2 - x_1$　$y_2 - y_1$

e.g.：$P(4, 0, 2)$，$Q(6, -1, 2)$

求 $\vec{PQ}$

$\vec{PQ} = (2, -1, 0) = \sqrt{5}$

向量有大小無方向

向量相加：若 $\vec{a} = [a_1, a_2, a_3]$

$\vec{b} = [b_1, b_2, b_3]$

$\vec{a} + \vec{b} = \vec{b} + \vec{a}$

實數與向量相乘有分配率

e.g.：$\vec{a} = [4, 0, 1]$，$\vec{b} = [2, 5, \frac{1}{3}]$

$\vec{a} + \vec{b} = [6, 5, \frac{4}{3}]$

單位向量大小為 1，有特定方向 $\langle \hat{i}, \hat{j}, \hat{k} \rangle$

$\vec{OP} = x\hat{i} + y\hat{j} + z\hat{k}$

e.g., $\vec{a} = 4\hat{i} + \hat{k}$

$\vec{b} = 2\hat{i} - 5\hat{j} + \frac{1}{3}\hat{k}$

$\vec{a} + \vec{b} = 6\hat{i} - 5\hat{j} + \frac{4}{3}\hat{k}$

$= [6, -5, \frac{4}{3}]$

Inner Product (內積)

定義: $\vec{a}, \vec{b}$ 內積 $\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta$

$\vec{a}\cdot\vec{b} = [a_1, a_2, a_3]\cdot[b_1, b_2, b_3]$

$= (a_1b_1, a_2b_2, a_3b_3) = a_1b_1 + a_2b_2 + a_3b_3$

$\vec{a}\cdot\vec{b} = 0$ 表正交(垂直)

$\cos\theta = \dfrac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$ / $|\vec{a}||\vec{b}| \neq 0$

e.g $\vec{a} = [1, 2, 0]$, $\vec{b} = [3, -2, 1]$

$\vec{a}\cdot\vec{b} = 3 - 4 + 0 = -1$

$|\vec{a}| = \sqrt{5}$

$|\vec{b}| = \sqrt{14}$

內積並滿足分配率

若一向量 $\vec{b} = |\vec{a}|\cos\theta$

則稱 $\vec{b}$ 為 $\vec{a}$ 在 $\vec{b}$ 之投影單位向量

外積 cross

定義 $\vec{v}$ 為一向量, $\vec{a}, \vec{b}$ 為另外兩個

若 $\vec{v} = \vec{a}\times\vec{b}$, 則 $\vec{v}$ 為 $\vec{a}$ 與 $\vec{b}$ 的外積

$|\vec{v}| = |\vec{a}\times\vec{b}| = |\vec{a}||\vec{b}|\sin\theta$

方向為 $\vec{a}$ & $\vec{b}$ 形成之平面

$\vec{v} = \vec{a}\times\vec{b}$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$= \hat{i}\begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \hat{j}\begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \hat{k}\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}$

$= (a_2b_3 - a_3b_2)\hat{i} - \cdots = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$

外積具備方向性 $\hat{i}\times\hat{j} = -\hat{j}\times\hat{i}$

e.g $\vec{a} = [1, 1, 0]$, $\vec{b} = [3, 0, 0]$

求 $\vec{v} = \vec{a}\times\vec{b}$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 0 \\ 3 & 0 & 0 \end{vmatrix} = 0\hat{i} + 0\hat{j} - 3\hat{k}$$

$\vec{a}\cdot(\vec{b}\times\vec{c})=$

一個由 $\vec{a},\vec{b},\vec{c}$ 所成的平行四面體的體積

向量微分

若 $\vec{V}=\vec{V}(P)=[V_1(P_1),\vec{V}_2(P_2),\vec{V}_3(P_3)]$

一無窮數列 a_n ($n=1,2,3\cdots$)

若 $\lim_{n\to\infty}|a_n-a|=0 \Rightarrow \lim_{n\to\infty} a_n=a$

收斂

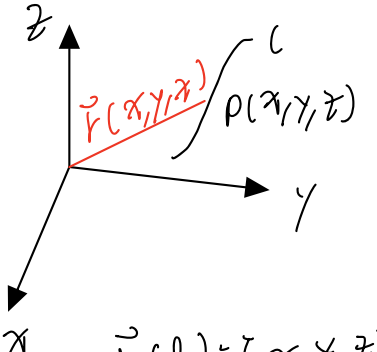
$\lim_{t\to t_0}|\vec{V}(t)-L|=0 \Rightarrow \lim_{t\to 0}\vec{V}(t)=L$

$\therefore \lim_{t\to t_0}\vec{V}(t)=\vec{V}(t_0) \Rightarrow$ 連續

向量微積分

$\vec{V}=[V_1,V_2,V_3]=V_1\hat{i}+V_2\hat{j}+V_3\hat{k}$

$$\frac{\partial\vec{V}}{\partial t}=\frac{\partial V_1}{\partial t}\hat{i}+\frac{\partial V_2}{\partial t}\hat{j}+\frac{\partial V_3}{\partial t}\hat{k}$$



$\vec{r}(t)=[x,y,z]$

$\therefore$ 一曲線的切線

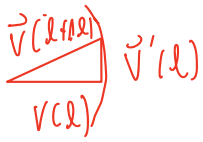
$$\vec{r}'(t)=\lim_{\Delta t\to 0}\frac{\vec{r}(t+\Delta t)-\vec{r}(t)}{\Delta t}$$

曲線長度：$l=\int_a^b\sqrt{\vec{r}'\cdot\vec{r}'}\,dt$

若向量函數可微分 $\Rightarrow$ 極限存在

$$\therefore \vec{V}(t)=\lim_{\Delta t\to 0}\frac{\vec{V}(t+\Delta t)-\vec{V}(t)}{\Delta t}$$

稱 $\vec{V}(t)'$ 為 $\vec{V}(t)$ 之導數



Gradients of scalar field

梯度

The setting is that we are given a scalar function $f(x,y,z)$ that is defined and differentiable in a domain in 3-space with cartesian coordinates x, y, z we denote the gradient of that function by grad f or $\vec{\nabla} f$

The gradient of $f(x,y,z)$

$$\text{grad } f = \vec{\nabla} f = \left[\frac{\partial f(x,y,z)}{\partial x}, \frac{\partial f(x,y,z)}{\partial y}, \frac{\partial f(x,y,z)}{\partial z}\right]$$

$$= \frac{\partial f(x,y,z)}{\partial x}\hat{i} + \frac{\partial f(x,y,z)}{\partial y}\hat{j} + \frac{\partial f(x,y,z)}{\partial z}\hat{k}$$

$$\vec{\nabla} = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}$$

Directional Derivative

$D_a f$ or $\frac{df}{ds}$ ← 代表一個 $f(x,y,z)$ 函數在向量 $\vec{a}$ 的方向上的一個點 P 的表示

$$\Rightarrow D_a f = \frac{df}{ds} = \lim_{s \to 0} \frac{f(a) - f(p)}{s}$$

$$\Rightarrow D_a f = \frac{df}{ds} = \vec{a} \cdot \text{grad } f$$

↳ $\vec{a}$ 為單位向量

e.g. $f(x,y,z) = 2x^2 + 3y^2 + z^2$

$P(2,1,3)$ at $\vec{a} = [1,0,-2]$ 方向

$$\text{grad } f = \frac{\partial f(x,y,z)}{\partial x}\hat{i} + \frac{\partial f(x,y,z)}{\partial y}\hat{j} + \frac{\partial f(x,y,z)}{\partial z}\hat{k}$$

$$= [4x, 6y, 2z]$$

$$\vec{\nabla} f(p) = (8, 6, 6)$$

$$|\vec{a}| = \sqrt{5}$$

$$\therefore D_a f = \frac{[1,0,-2]}{\sqrt{5}} \cdot [8,6,6] = \frac{1}{\sqrt{5}}(8 + 0 - 12)$$

$$= \frac{-4}{\sqrt{5}}$$

定理 gradient as surface Normal vector

法向量

Let $f(x,y,z)$ be a differentiable scalar in space $\Rightarrow f(x,y,z) = C$, C is constant represent a surface S. Then if the gradient of $f(x,y,z)$ at point p of S is the Zero vector, it is a normal vector of S at P

e.g. Find a unit vector $\hat{n}$ of the 錐體 cone of recolution $z^2 = 4(x^2+y^2)$ at $P(1,0,2)$

$$\vec{u} \frac{[8,0,-4]}{|\text{grad } f(p)|} = \left[\frac{2}{\sqrt{5}}, 0, \frac{-1}{\sqrt{5}}\right]$$

$f = 0$, $f(x,y,z) = 4(x^2+y^2) - z^2$

$\text{grad } f = [8x, 8y, -2z]$

$\text{grad } f(P) = [8, 0, -4]$

張秋怡

divergence of a vector field

散度

$\vec{V}(x,y,z) \Rightarrow \vec{V} = [V_1(x,y,z),\ V_2(x,y,z),\ V_3(x,y,z)] = [V_1(x,y,z)\hat{i} + V_2(x,y,z)\hat{j} + V_3(x,y,z)\hat{k}]$

$$\text{div}\,\vec{V} = \frac{\partial V_1(x,y,z)}{\partial x} + \frac{\partial V_2(x,y,z)}{\partial y} + \frac{\partial V_3(x,y,z)}{\partial z}$$

$$\Rightarrow \text{div}\,\vec{V} = \vec{\nabla}\cdot\vec{V} = \left[\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right]\cdot[V_1(x,y,z), V_2(x,y,z), V_3(x,y,z)]$$

$$= \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\cdot\left(V_1(x,y,z)\hat{i} + V_2(x,y,z)\hat{j} + V_3(x,y,z)\hat{k}\right) = \bigcirc$$

$$\nabla = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}$$

∇^2 = Laplacian operator 拉普拉斯操作子

for $\vec{V}$: grad f

$$= \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right]$$

$$= \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k}$$

$$\text{div}\,\vec{V} = \text{div}(\text{grad}f) = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\cdot\left(\frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k}\right)$$

$$= \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = \nabla^2 f$$

the divergence of the gradient is the Laplacian

curl of a vector field

旋度

$\vec{V}(x,y,z) = [V_1(x,y,z), V_2(x,y,z), V_3(x,y,z)]$

$= V_1(x,y,z)\hat{i} + V_2(x,y,z)\hat{j} + V_3(x,y,z)\hat{k}$

$$\text{curl}\,\vec{V} = \vec{\nabla}\times\vec{V} = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\times\left[V_1(x,y,z)\hat{i} + V_2(x,y,z)\hat{j} + V_3(x,y,z)\hat{k}\right]$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_1(x,y,z) & V_2(x,y,z) & V_3(x,y,z) \end{vmatrix} = \bigcirc\hat{i} - \bigcirc\hat{j} + \bigcirc\hat{k}$$

e.g

curl of a vector function

Let $\vec{v}=[yz, 3zx, z]=yz\hat{i}+3zx\hat{j}+z\hat{k}$

$\text{curl}\,\vec{v}=\left(\frac{\partial}{\partial x}\hat{i}+\frac{\partial}{\partial y}\hat{j}+\frac{\partial}{\partial z}\hat{k}\right)\times\left(yz\hat{i}+3zx\hat{j}+z\hat{k}\right)$

$$=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\\ yz & 3zx & z\end{vmatrix}=(0-3x)\hat{i}-(0-y)\hat{j}+(3z-z)\hat{k}$$

$$=-3x\hat{i}+y\hat{j}+2z\hat{k}$$

定理：curl(grad f)=0 ⇐ 如果一個連續可微的向量場是純量函數的梯度則其旋度為0

$\because \text{grad}\, f=\left(\frac{\partial f}{\partial x}\hat{i}+\frac{\partial f}{\partial y}\hat{j}+\frac{\partial f}{\partial z}\hat{k}\right)$，curl(grad f) $\Rightarrow \nabla\times(\text{grad}\, f)$

$$=\left(\frac{\partial}{\partial x}\hat{i}+\frac{\partial}{\partial y}\hat{j}+\frac{\partial}{\partial z}\hat{k}\right)\times\left(\frac{\partial f}{\partial x}\hat{i}+\frac{\partial f}{\partial y}\hat{j}+\frac{\partial f}{\partial z}\hat{k}\right)$$

$$\text{curl}(\text{grad}\, f)=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial}{\partial z}\end{vmatrix}=0$$

張秀宜

§向量積分與定理

12.1

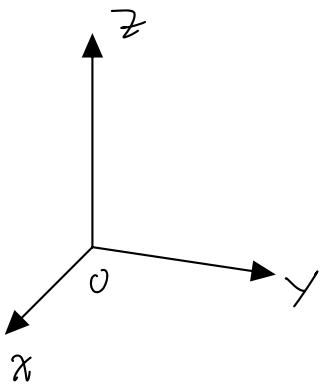
Line integral (與路徑無關，起點，終點有關

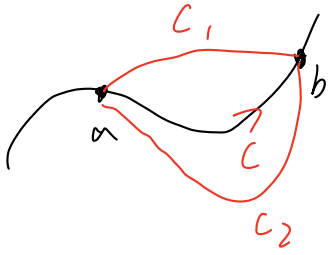
$\int_a^b f(x)\,dx$

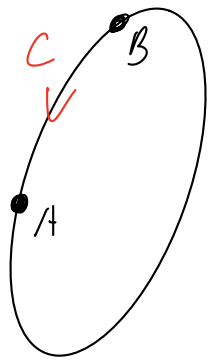
① 若 $\vec{F}=\text{grad}f$，其中 f 為任意函數

② 若在一封閉曲線中 $\oint_C \vec{F}\cdot d\vec{r}=0$

③ 若 $\vec{F}$ 向量函數，其旋度為0，$\Rightarrow \vec{F}\times\vec{F}=\vec{0}$







close Loop

Path of integral $\quad d\vec{r}=\frac{d\vec{r}}{dt}dt=\vec{r}'dt$

$\Rightarrow \int_C \vec{F}(r)\cdot d\vec{r}=\int_a^b \vec{F}(r(t))\cdot\vec{r}(t)\,dt$，其中 $\vec{r}(t)=\frac{d\vec{r}}{dt}$

$\Rightarrow \int_C \vec{F}(r)\cdot d\vec{r}=\int_C(\vec{F}_1\cdot d\vec{x}+\vec{F}_2\cdot d\vec{y}+\vec{F}_3\cdot d\vec{z})$

$=\int_a^b(F_1x'(t)+F_2y'(t)+F_3z'(t))\,dt$ ← 也為線積分

e.g

$\vec{F}=[2x,2y,4z]$，$f=x^2+y^2+z^2$

$\Rightarrow \because \text{grad}f=[2x,2y,4z]$

$\therefore$ 若 $A(1,1,1)$，$B(0,0,0)$

則

$f(A)-f(B)=16$

若有一路徑 $C=r(t)$

$=[t,t,t]$

$0\le t\le 2$

$\vec{F}(r(t))=[2t,2t,4t]$

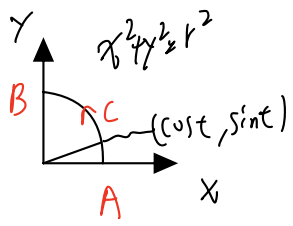
$\therefore \vec{F}(r(x))\cdot\vec{r}'(t)$

$=2t+2t+4t=8t$

$\int_C \vec{F}(r(t))\cdot\vec{r}'(t)\,dt$

$=\int_0^2 8t\,dt=4t^2\Big|_0^2=16$

e.g



$\vec{F}=[-y,-xy]$

$\vec{r}(t)=[\cos t,\sin t]$

$=\cos t\,\hat{i}+\sin t\,\hat{j}$

$0\le t\le\frac{\pi}{2}$

$x(t)=\cos t$

$y(t)=\sin t$

$\therefore \vec{F}(r(t))=-y(t)\hat{i}-x(t)y(t)\hat{j}$

$=[-\sin t,\cos t\sin t]$

$=-\sin t\,\hat{i}-\cos t\sin t\,\hat{j}$

$$\therefore \int_C \vec{F}(\vec{r})\,d\vec{r} = \int_0^{\frac{\pi}{2}} [-\sin t - \cos t\sin t]\cdot[-\sin t, \cos t]\,dt$$

$$= \int_0^{\frac{\pi}{2}} (\sin^2 t - \cos^2 t \sin t)\,dt$$

$$= \int_0^{\frac{\pi}{2}} \sin^2 t\,dt - \int_0^{\frac{\pi}{2}} \cos^2 t \sin t\,dt = \frac{\pi}{4} - \frac{1}{3} \approx 0.4521$$

$$\int_C (\vec{F} + \vec{G})\cdot d\vec{r} = \int_C \vec{F}\cdot d\vec{r} + \int_C \vec{G}\cdot d\vec{r}$$

$$\int_C \vec{F}\cdot d\vec{r} = \int_{C_1} \vec{F}\cdot d\vec{r} + \int_{C_2} \vec{F}\cdot d\vec{r}$$

定理：在 C 上給定任何相同的方向，
其產生的線積分的值不同

pf : $a \le t \le b$; $a^* \le t^* \le b^*$

Green Theorem 格林定理

$$\iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) \underbrace{dx\,dy}_{dA} = \oint_C (F_1\,dx + F_2\,dy)$$

$$\iint_R (\vec{\nabla} \times \vec{F})\cdot \vec{k}\,dx\,dy = \oint \vec{F}\cdot d\vec{r}$$

e.g

$F_1 = y^2 - 7y$; $F_2 = 2xy + 2x$, C 為 $x^2 + y^2 = 1$

prf Green Theorem

(1) $\because \iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) dx\,dy = \iint_R [2y + 2 - (2y - 7)]\,dx\,dy$

$$= 9 \iint_R dx\,dy = 9\pi$$

(2) $F_1 = x^2 - 7y = \sin^2 t - 7\sin t$; $F_2 = 2xy + 2x = 2\cos t \sin t + 2\cos t$

$$\oint_C (F_1 x' + F_2 y')\,dt$$

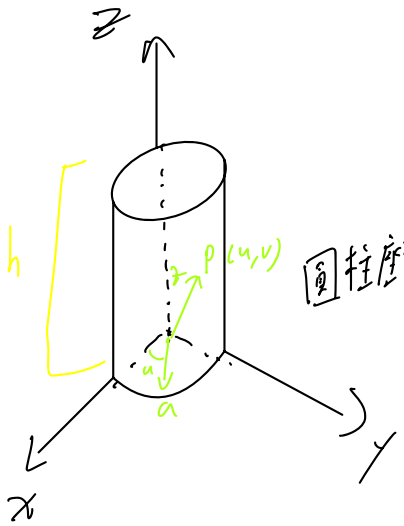
$$= \int_0^{2\pi} [\sin^2 t - 7\sin t](-\sin t) + 2(\cos t \sin t + 2\cos t)\,dt$$

$$= 7\pi + 2\pi = 9\pi$$

(x, y, z)

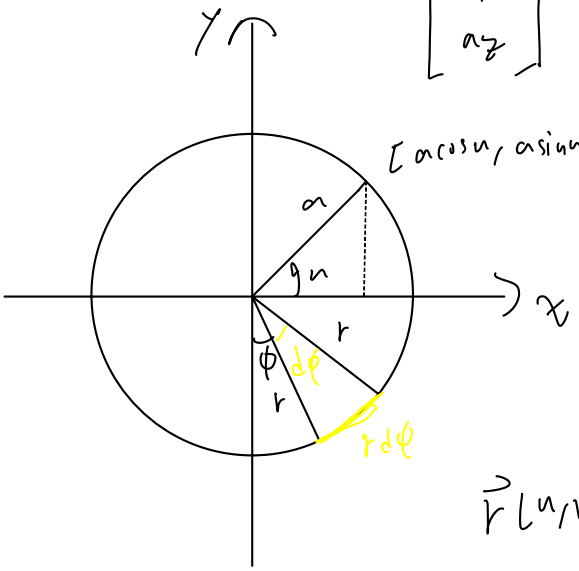
$\hat{i}, \hat{j}, \hat{k}$ ⇒ 直角座標的單位向量

$$\begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_r \\ a_\phi \\ a_z \end{bmatrix}$$



圓柱座標 (r, ϕ, z)

$\vec{r} = |\vec{r}| \cdot \hat{r}$



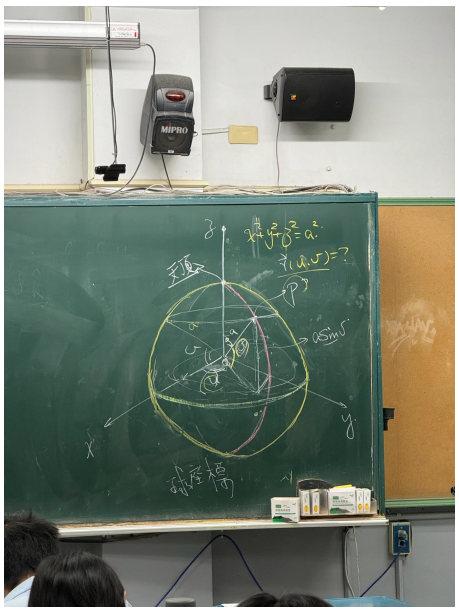
$[a\cos u, a\sin u]$

u 座標

$\vec{r}(u,v) = [a\cos u, a\sin u]$

$\vec{r}(u,v) = [x(u,v), y(u,v), z(u,v)]$

$= x(u,v)\hat{i} + y(u,v)\hat{j} + z(u,v)\hat{k}$



$\vec{r}(u,v) = a\cos v\cos u\,\hat{i} + a\cos v\sin u\,\hat{j} + a\sin v\,\hat{k}$

$0 \le u \le 2\pi, \ -\frac{\pi}{2} \le v \le \frac{\pi}{2}$

$x = a\cos v\sin u, \ y = a\cos v\sin u, \ z = a\sin v\,\hat{k}$

球座標與直角座標差距離

$0 \le \theta \le \pi$ 為天頂

$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$\because x = r\sin\theta\cos\phi; \ y = r\sin\theta\sin\phi; z = r\cos\theta$

代入位置向量

$\vec{r} = r\sin\theta\cos\phi\,\hat{i} + r\sin\theta\sin\phi\,\hat{j} + r\cos\theta\,\hat{k}, \ \vec{r} = \vec{r}(r, \phi, \theta)$

對 r 偏微 $\frac{\partial \vec{r}}{\partial r} = \frac{\partial x}{\partial r}\hat{i} + \frac{\partial y}{\partial r}\hat{j} + \frac{\partial z}{\partial r}\hat{k}$

$\frac{\partial \vec{r}}{\partial r} = \sin\theta\cos\phi\,\hat{i} + \sin\theta\sin\phi\,\hat{j} + \cos\theta\,\hat{k}$ — ⓐ

$\frac{\partial \vec{r}}{\partial \theta} = \frac{\partial x}{\partial \theta}\hat{i} + \frac{\partial y}{\partial \theta}\hat{j} + \frac{\partial z}{\partial \theta}\hat{k}$

$\frac{\partial \vec{r}}{\partial \theta} = r\cos\theta\cos\phi\,\hat{i} + r\cos\theta\sin\phi\,\hat{j} - r\sin\theta\,\hat{k}$ — ⓑ

$\frac{\partial \vec{r}}{\partial \phi} = -r\sin\theta\sin\phi\,\hat{i} + r\sin\theta\cos\phi\,\hat{j}$ — ⓒ

未知簡單位向量 $\hat{\theta}$

$$\left|\frac{\partial \vec{r}}{\partial \theta}\right| = \sqrt{r^2\cos^2\theta\cos^2\phi + r^2\cos^2\theta\sin^2\phi + r^2\sin^2\theta}$$

$\therefore$ 將 $\frac{\partial r}{\partial \theta}$

$$= \sqrt{r^2\cos^2\theta(\cos^2\phi+\sin^2\phi) + r^2\sin^2\theta}$$

$$= \sqrt{r^2\cos^2\theta + r^2\sin\theta}$$

$$= \sqrt{r^2[\cos^2\theta + \sin^2\theta)}$$

$$= r$$

$$\begin{bmatrix}\hat{r}\\ \hat{\theta}\\ \hat{\phi}\end{bmatrix} = \begin{bmatrix}\sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta\\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta\\ -\sin\phi & \cos\phi & 0\end{bmatrix}\begin{bmatrix}\hat{i}\\ \hat{j}\\ \hat{k}\end{bmatrix}$$

Surface intergral

(面積分)

1) 在 x, y, z 空間中

$z = \sqrt{a^2-x^2-y^2}$

$z^2 = a^2-x^2-y^2 \Rightarrow z^2+x^2+y^2=a^2$

e.g Trasformation of Double intergral of the Laplacian of a function into a line intergral of its Normal Derivate

Laplacian $\nabla^2 f$

法線導數

Soli:

$\because$ 函數 $f \Rightarrow$ 設 $F_1 = \frac{\partial f}{\partial y}$; $F_2 = \frac{\partial f}{\partial x}$

$$\Rightarrow \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = \nabla^2 f \quad \cdots ⓐ$$

$\therefore \oint_C (F_1 dx + F_2 dy) = \oint_C \left(F_1\frac{dx}{ds} + F_2\frac{dy}{ds}\right) ds$

$$= \oint_C \left(-\frac{\partial f}{\partial y}\frac{dx}{ds} + \frac{\partial f}{\partial x}\frac{dy}{ds}\right) ds \quad \cdots ⓑ$$

s 為 C 的弧長

$\because$ $\text{grad} f \cdot \hat{n} = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right] \cdot \left[\frac{dy}{ds}, -\frac{dx}{ds}\right]$

C 為 C 的單位法向量

$$= \frac{\partial f}{\partial x}\frac{dy}{ds} - \frac{\partial f}{\partial y}\frac{dx}{ds}$$

$\therefore \vec{r}'(s) = \frac{d\vec{r}}{ds} = \left[\frac{dx}{ds}, \frac{dy}{ds}\right]$

normal dirivate of f

$\Rightarrow \frac{\partial f}{\partial n} = (\text{grad}) \cdot \hat{n}$

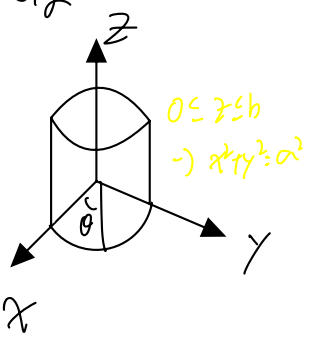
Triple integral (Divergence theorem of Gauss)

$\because \vec{F} = [F_1, F_2, F_3]$

$= F_1 \hat{i} + F_2 \hat{j} + F_3 \hat{k}$

$\operatorname{div} \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$

e.g.



$\vec{F} = [x^1, xy, x^2z]$

$\operatorname{div} \vec{F} = 3x^2 + x + x^2 = 5x^2$

$x = a\cos\theta, \quad y = a\sin\theta$

$\therefore dx\,dy\,dz = r\,dr\,d\theta\,dz$

$\therefore \iiint_T (\operatorname{div} \vec{F})\,dV = \iiint 5x^2 \cdot r\,dr\,d\theta\,dz$

$\int_0^{2\pi} \int$

定理 1

$\iiint_T \operatorname{div} \vec{F}\,dV = \iint_S \vec{F} \cdot \hat{n}\,dA$

$dx\,dy\,dz$

流量 Flux

$\hat{n} = [\cos\alpha, \cos\beta, \cos\gamma]$

$\iiint_T \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \right) dx\,dy\,dz$

$= \iint_S (F_1 \cos\alpha + F_2 \cos\beta + F_3 \cos\gamma)\,dA$

$= \iint_S (F_1\,dy\,dz + F_2\,dz\,dx + F_3\,dx\,dy)$

10.9

Stoke's Theorem

:) 表面積 :) 線積分，線積分 :) 面積分

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} \qquad \iint_S (\operatorname{curl}\vec{F})\,\hat{n}\,dA = \oint_C \vec{F}\cdot\vec{r}\,'(s)\,ds$$

e.g.

$z = f(x,y) = 1-(x^2+y^2)$, $z \geq 0$

Sol:

$\vec{r}(s) = [-\sin s, \cos s, 0]$

$\vec{F}(r(s)) = [\sin s, 0, \cos s]$

$$\oint_C \vec{F}\cdot d\vec{r} = \int_0^{2\pi} \vec{F}(r)\,\vec{r}\,'(s)\,ds$$

$$= \int_0^{2\pi} [\sin s(-\sin s) + 0 + 0]\,ds$$

$$= -\pi$$

$\operatorname{curl}\vec{F} = \operatorname{curl}(F_1, F_2, F_3)$

$= \operatorname{curl}(y, z, x)$

$= (-1, -1, -1)$

$\vec{N}$: $\operatorname{grad}(z - f(x,y,s)) = (2x, 2y, 1)$

$\operatorname{curl}\vec{F}\cdot\vec{N} = -2x-2y-1$, $\hat{n}\,dA = \vec{N}\,dx\,dy$

$$\therefore \iint_S (\operatorname{curl}\vec{F})\,\hat{n}\,dA = \iint_R (\operatorname{curl}F)\cdot\vec{N}\,dx\,dy$$

$$= \iint_R (-2x-2y-1)\,dx\,dy$$

$$= \int_0^{2\pi}\int_0^1 (-2r(\cos\theta+\sin\theta)-1)$$

$$= -\pi$$

Chapter 7.

矩陣，行列式

A matrix is a rectangular array of number or function will be enclose in a brakets

e.g

$$\begin{bmatrix} 0.3 & 1 & -5 \\ 0 & -0.2 & 16 \end{bmatrix}$$

↓ 方矩陣

2) Square

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

若矩陣只有單一個行或列稱為 Vector (向量)

row vector (列向量)

column Vector (行向量)

$$\begin{bmatrix} 4 & 6 & 9 \\ 6 & 0 & -2 \\ 5 & -8 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 20 \\ 10 \end{bmatrix}$$

八

矩陣相等

$A=[a_{jk}]$ $B=[b_{jk}]$

if $A=B$

則其 size 要對應 vector 要相等

so that

$a_{11}=b_{11}$; $a_{12}=b_{12}$ ...

e.g

$$A=\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$B=\begin{bmatrix} 4 & 0 \\ 3 & -1 \end{bmatrix}$$

則 $a_{11}=4$, $a_{12}=0$ ...

liner system

x_1 x_2 x_3

↓ ↓ ↓

變數

$a_1x_1+a_2x_2+a_3x_3=b_1$ 方程式

$$\begin{cases} 4x_1+6x_2+9x_3=6 \\ 6x_1 \quad -2x_3=20 \\ 5x_1-8x_2+x_3=10 \end{cases}$$

$$B=\begin{bmatrix} 6 \\ 20 \\ 10 \end{bmatrix} \quad A=\begin{bmatrix} 4 & 6 & 9 \\ 6 & 0 & -2 \\ 5 & -8 & 1 \end{bmatrix} \quad x=\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$AX=B$

矩陣的相加

$A+B$ (size 須相等)

$$A=\begin{bmatrix} -4 & 6 & 3 \\ 0 & 1 & 2 \end{bmatrix}$$

$$B=\begin{bmatrix} 5 & -1 & 0 \\ 3 & 0 & 0 \end{bmatrix}$$

$$A+B=\begin{bmatrix} 1 & 5 & 3 \\ 3 & 1 & 2 \end{bmatrix}$$

矩陣乘上數字 C

$=CA$

規則

$A+B=B+A$

$(A+B)+C=A+(B+C)$

$A+0=A$

$A+(-A)=0$

矩陣相乘

A為 $m\times n$ 矩陣 $A=[a_{jk}]$

B為 $n\times r$ 矩陣 $B=[b_{jk}]$

$AB=C$

$[m\times n][n\times p]=[m\times p]$

$C(A+B)=CA+CB$

$(C+k)A=CA+kA$

$IA=A$

↑

單位矩陣

e.g

$$A=\begin{bmatrix}3 & 5 & -1\\ 4 & 0 & 2\\ -6 & -3 & 2\end{bmatrix},\quad B=\begin{bmatrix}2 & -2 & 3 & 1\\ 5 & 0 & 7 & 8\\ 9 & -4 & 1 & 1\end{bmatrix}$$

$$AB=\begin{bmatrix}22 & -2 & 43 & 42\\ 26 & -16 & 14 & 6\\ -9 & 4 & -17 & -18\end{bmatrix}$$

$AB \neq BA$

$[m\times n][r\times p]$ $[r\times p][m\times n]$

$A=A^T$

$[a_{jk}]=[a_{kj}]$

e.g

$$A=\begin{bmatrix}5 & -8 & 1\\ 4 & 0 & 0\end{bmatrix}$$

$$A^T=\begin{bmatrix}5 & 4\\ -8 & 0\\ 1 & 0\end{bmatrix}$$

The rules of transposition of matrix

$(A^T)^T=A$

$(A+B)^T=A^T+B^T$

$(CA)^T=CA^T$

$\boxed{(AB)^T=B^TA^T}$

特殊矩陣

(對稱)：一個方矩陣

$A^T=A$

$$A^T=A=\begin{bmatrix}1 & 2 & 3\\ 2 & 0 & 4\\ 3 & 4 & 0\end{bmatrix}$$

對角矩陣

(Diagonal matrix)

$$\begin{bmatrix}a_{11} & 0 & 0\\ 0 & a_{22} & 0\\ 0 & 0 & a_{33}\end{bmatrix}$$

a_{11},a_{22},a_{33} 均為0

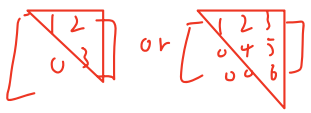
若皆為5且保持為純量矩陣

(scalar matrix)

若5為1則為I (unit matrix)

上三角與下三角矩陣

(upper and lower triangular matrix)



某一個元素均為0

e.g.

$2x_1 + 5x_2 = 2$

$-4x_1 + 3x_2 = -30$

Sol:

Coefficient matrix $A = \begin{bmatrix} 2 & 5 \\ -4 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 2 \\ -30 \end{bmatrix}$

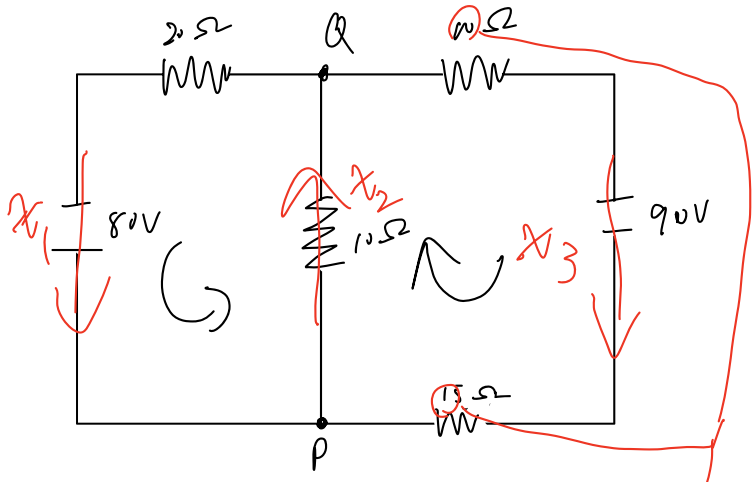
∴ 其擴張矩陣 (augmented matrix) $[A|B] = \left[\begin{array}{cc|c} 2 & 5 & 2 \\ -4 & 3 & -30 \end{array}\right]$ ⟲ ×2

$= \left[\begin{array}{cc|c} 2 & 5 & 2 \\ 0 & 13 & -26 \end{array}\right]$

$13x_2 = -26$

$x_2 = -2$

$x_1 = 6$



I_1, I_2, I_3

$\Rightarrow x_1, x_2, x_3$

$\begin{cases} x_1 - x_2 + x_3 = 0 \\ -x_1 + x_2 - x_3 = 0 \end{cases}$

$20x_1 + 10x_2 = 80$

$10x_2 + 25x_3 = 90$

$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ -1 & 1 & -1 & 0 \\ 20 & 10 & 0 & 80 \\ 0 & 10 & 25 & 90 \end{array}\right]$ ⟲ ×20

$\Rightarrow \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 30 & -20 & 80 \\ 0 & 10 & 25 & 90 \end{array}\right] \Rightarrow \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 30 & -20 & 80 \\ 0 & 10 & 25 & 90 \\ 0 & 0 & 0 & 0 \end{array}\right]$ ⟲ ×−3

$\Rightarrow \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 0 & -95 & -190 \\ 0 & 10 & 25 & 90 \\ 0 & 0 & 0 & 0 \end{array}\right]$

$95x_3 = 190$

$x_3 = 2$

$10x_2 = 40$

$x_2 = 4$

$x_1 = 2$

$$\left[\begin{array}{cccc|c} 3 & 2 & 2 & -5 & 8 \\ 0.6 & 1.5 & 1.5 & -5.4 & 2.7 \\ 1.2 & -0.3 & -0.3 & 2.4 & 2.1 \end{array}\right] \begin{matrix} \\ -0.2 \\ -0.4 \end{matrix}$$

$r_2 = r_3$

$$= \left[\begin{array}{cccc|c} 3 & 2 & 2 & -5 & 8 \\ 0 & 1.1 & 1.1 & -4.4 & 1.1 \\ 0 & -1.1 & -1.1 & 4.4 & -1.1 \end{array}\right] \begin{matrix} \\ \\ \div -1 \end{matrix}$$

$$= \left[\begin{array}{cccc|c} 3 & 2 & 2 & -5 & 8 \\ 0 & 1.1 & 1.1 & -4.4 & 1.1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

無限多組解

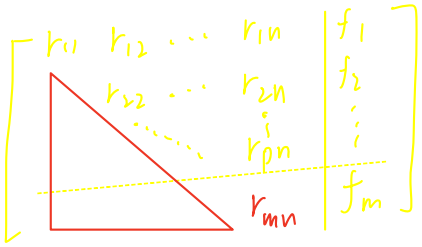
$$\left[\begin{array}{ccc|c} 3 & 2 & 1 & 3 \\ 2 & 1 & 1 & 0 \\ 6 & 2 & 4 & 6 \end{array}\right] \begin{matrix} \\ -\frac{1}{3} \\ \end{matrix} \begin{matrix} \\ \\ \div 2 \end{matrix}$$

$$= \left[\begin{array}{ccc|c} 6 & 2 & 4 & 6 \\ 0 & \frac{1}{3} & -\frac{1}{3} & -2 \\ 0 & 1 & -1 & 0 \end{array}\right] \begin{matrix} \\ \\ \div 3 \end{matrix}$$

$$= \left[\begin{array}{ccc|c} 6 & 2 & 4 & 6 \\ 0 & \frac{1}{3} & -\frac{1}{3} & -2 \\ 0 & 0 & 0 & -6 \end{array}\right]$$

$0 = -6$

無解



$p \le m$, $r_{11} \neq 0$

紅色部份都為0

高斯消去法最簡表示

if $p < m$, 只要任一項為0則無解

若 $p = m = n$ (方形矩陣), 則有唯一解

若 $p = n$, $n \neq m$, 則有無限多組解

linear independent, Rank of matrix

線性獨立　　秩

$c_1\alpha_1 + c_2\alpha_2 + \cdots + c_n\alpha_n$

若此式為0，則無數個解

若有一不全為0之解，代表基線性組合為0，則稱為 linear dependent (線性相依)

Rank of Matrix

定義：矩陣的 Rank 為基線性獨立列向量的最大數，寫成 RankA

若 A = 0，則 RankA = 0

e.g

$$\left[\begin{array}{ccc|c} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ 21 & -21 & 0 & 15 \end{array}\right]$$

$$\Rightarrow \left[\begin{array}{ccc|c} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ 0 & -21 & -14 & 1 \end{array}\right] = \left[\begin{array}{ccc|c} 3 & 0 & -2 & 2 \\ 0 & 42 & 28 & 58 \\ 0 & 0 & 0 & 30 \end{array}\right]$$

RankA = 2

定理：一個矩陣A的Rank

等於基向量線性獨立的最大數

因此，矩陣A與其轉置

$\text{Rank}A^T = \text{Rank}A$

h.w

e.g $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{bmatrix}$

$\Rightarrow \text{Rank}A = 2$; $\text{Rank}A^T = 2$

線性系統的解的有無與唯一

三元一個方程組有n個未知數，只有一個唯一解

如果係數矩陣與擴張矩陣有相同的

① Rank A = n

Rank A = n ; Rank $\widehat{A}$ = n

四若一般係數矩陣 Rank A < Rank $\widehat{A}$，則沒無限多組解

Determinant (行列式)

n階行列式為一個 n×n 矩陣

$A = [a_{ij}]$

$D = \det A = \begin{vmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{vmatrix}$

對於 n=1, D = def A = a_{11}

n≥2 / $D = a_{j1}C_{j1} + a_{j2}C_{j2} + \cdots + a_{jn}C_{jn}$ (j = 1 ... n)

or

$D = a_{1k}C_{1k} + a_{2k}C_{2k} + \cdots a_{nk}C_{nk}$ (k = 1, 2, ... n)

$C_{jk} = (-1)^{j+k} M_{jk}$ ...

其中 M_{jk} ... Minor of a_{jk} in D ; C_{jk} ... the cofaction of a_{jk} in D

$D = \sum_{k=1}^{n} (-1)^{j+k} a_{jk} M_{jk}$ (j = 1, 2 ... n)

$D = \sum_{j=1}^{n} (-1)^{j+1} a_{jk} M_{jk}$ (k = 1, 2, ... n)

符號 $\begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix}$

e.g. $D = \begin{vmatrix} 1 & 3 & 0 \\ 2 & 6 & 4 \\ -1 & 0 & 2 \end{vmatrix}$ ⇒ 3×3

⇒ $D = 1 \times \begin{vmatrix} 6 & 4 \\ 0 & 2 \end{vmatrix} - 3 \times \begin{vmatrix} 2 & 4 \\ -1 & 2 \end{vmatrix} + 0 \times \begin{vmatrix} 2 & 6 \\ -1 & 0 \end{vmatrix}$

= 12 − 24 = −12 ##

克拉默公式

设 $\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1 \\ a_{21}x_1 + a_{22}x_2 = b_2 \end{cases}$

$\Rightarrow x_1 = \dfrac{\begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}}{\det A}$; $x_2 = \dfrac{\begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}}{\det A}$

$= \dfrac{b_1 a_{22} - a_{12} b_2}{\det A}$ $= \dfrac{a_{11} b_2 - b_1 a_{21}}{\det A}$

$\det A \neq 0$

同理3阶 $\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3 \end{cases}$

$x_1 = \dfrac{\begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix}}{\det A}$ $\Rightarrow \det D$ ...

定理：n阶行列式的性质

(a) 二个列或比互换，行列式的值乘(-1)

(b) 加任一列的值到某列，行列式之值

(c) 若某一列乘上 C 则其值乘 C

Inverse of a matrix

$A \cdot A^{-1} = A^{-1} \cdot A = I$

若A有一个Inverse且inverse为unique

若A有一个Inverse，A为nonsingular matrix

若A没有Inverse则称singular matrix

若B，C为A的Inverse

则 $AB = I$，$CA = I$; $B = IB = (CA)B$

$= CI = C$

求反矩陣

例 $A=\begin{bmatrix}-1 & 1 & 2\\ 3 & -1 & 1\\ -1 & 3 & 4\end{bmatrix}$，求 $A^{-1}=?$

Sol: $[AI]=\left[\begin{array}{ccc|ccc}-1 & 1 & 2 & 1 & 0 & 0\\ 3 & -1 & 1 & 0 & 1 & 0\\ -1 & 3 & 4 & 0 & 0 & 1\end{array}\right]\Rightarrow\left[\begin{array}{ccc|ccc}-1 & 1 & 2 & 1 & 0 & 0\\ 0 & 2 & 7 & 3 & 1 & 0\\ 0 & 2 & 2 & -1 & 0 & 1\end{array}\right]\Rightarrow\left[\begin{array}{ccc|ccc}-1 & 1 & 2 & 1 & 0 & 0\\ 0 & 2 & 7 & 3 & 1 & 0\\ 0 & 0 & -5 & -4 & -1 & 1\end{array}\right]$

轉成單位矩陣

$\Rightarrow\left[\begin{array}{ccc|ccc}1 & -1 & 2 & -1 & 0 & 0\\ 0 & 1 & 3.5 & 1.5 & 0.5 & 0\\ 0 & 0 & 1 & 0.8 & 0.2 & -0.2\end{array}\right]\Rightarrow\left[\begin{array}{ccc|ccc}1 & -1 & 0 & 0.6 & 0.9 & -0.4\\ 0 & 1 & 0 & -1.3 & -0.2 & 0.7\\ 0 & 0 & 1 & 0.8 & 0.2 & -0.2\end{array}\right]\Rightarrow\left[\begin{array}{ccc|ccc}1 & 0 & 0 & -0.7 & 0.2 & 0.3\\ 0 & 1 & 0 & -1.3 & -0.2 & 0.7\\ 0 & 0 & 1 & 0.8 & 0.2 & -0.2\end{array}\right]$

I A^{-1}

$[AI]=\left[\begin{array}{ccc|ccc}-1 & 1 & 2 & 1 & 0 & 0\\ 3 & -1 & 1 & 0 & 1 & 0\\ -1 & 3 & 4 & 0 & 0 & 1\end{array}\right]\Rightarrow\left[\begin{array}{ccc|ccc}1 & -1 & -2 & -1 & 0 & 0\\ 0 & 2 & 7 & 3 & 1 & 0\\ 0 & 2 & 2 & -1 & 0 & 1\end{array}\right]\Rightarrow\left[\begin{array}{ccc|ccc}1 & -1 & -2 & -1 & 0 & 0\\ 0 & 2 & 7 & 3 & 1 & 0\\ 0 & 0 & -5 & -4 & -1 & 1\end{array}\right]$

$\Rightarrow\left[\begin{array}{ccc|ccc}1 & -1 & -2 & -1 & 0 & 0\\ 0 & 1 & 3.5 & 1.5 & 0.5 & 0\\ 0 & 0 & 1 & 0.8 & 0.2 & -0.2\end{array}\right]\Rightarrow\left[\begin{array}{ccc|ccc}1 & -1 & -2 & -1 & 0 & 0\\ 0 & 1 & 0 & -1.3 & -0.2 & 0.7\\ 0 & 0 & 1 & 0.8 & 0.2 & -0.2\end{array}\right]\Rightarrow\left[\begin{array}{ccc|ccc}1 & 0 & 0 & -0.7 & 0.2 & 0.3\\ 0 & 1 & 0 & -1.3 & -0.2 & 0.7\\ 0 & 0 & 1 & 0.8 & 0.2 & -0.2\end{array}\right]$

$3.5\times 0.8=2.8$

$-2.3\times 1.6=-0.2$